1. Python Basics

- Q: What is Python?
  A: Python is a high-level, interpreted, object-oriented programming language known for readability and vast libraries.
- Q: What are Python’s key features?
  A: Easy syntax, dynamic typing, interpreted, cross-platform, large community support.
- Q: What is dynamic typing?
  A: Variable type is determined at runtime.
- Q: Difference between list and tuple?
  A: List is mutable; tuple is immutable.
- Q: What is PEP 8?
  A: Official Python style guide for writing clean and readable code.
- Q: What is shallow copy vs deep copy?
  A: Shallow copy copies references; deep copy copies nested objects as well.

2. OOP in Python

- Q: What is OOP?
  A: Programming paradigm based on objects and classes.
- Q: Four pillars of OOP?
  A: Encapsulation, Abstraction, Inheritance, Polymorphism.
- Q: What is __init__?
  A: Constructor method executed when object is created.
- Q: What is method overriding?
  A: Redefining parent method in child class.
- Q: What is super()?
  A: Used to call parent class methods.
- Q: What is abstraction?
  A: Hiding implementation details using abstract classes.

3. Advanced Python

- Q: What are decorators?
  A: Functions that modify behavior of another function.
- Q: What are generators?
  A: Functions using yield to produce values lazily.
- Q: What is GIL?
  A: Global Interpreter Lock allows only one thread to execute Python bytecode at a time.
- Q: What is lambda?
  A: Anonymous single-expression function.
- Q: What is list comprehension?
  A: Short syntax to create lists in one line.
- Q: What is a context manager?
  A: Manages resources using 'with' statement.

4. Data Structures & Algorithms

- Q: What is time complexity?
  A: Measure of algorithm efficiency using Big-O notation.
- Q: Explain O(1), O(n), O(log n).
  A: Constant, Linear, Logarithmic time complexities.
- Q: What is a stack?
  A: LIFO data structure.
- Q: What is a queue?
  A: FIFO data structure.
- Q: What is binary search?
  A: Efficient search algorithm for sorted data (O(log n)).
- Q: Difference between array and linked list?
  A: Array uses contiguous memory; linked list uses nodes with pointers.

5. Flask Interview

- Q: What is Flask?
  A: Lightweight Python web framework.
- Q: What is @app.route?
  A: Decorator mapping URL to function.
- Q: Difference between GET and POST?
  A: GET retrieves data; POST sends data securely.
- Q: What is Jinja2?
  A: Flask's templating engine.
- Q: What is SQLAlchemy?
  A: ORM for database interaction.

6. Django Interview

- Q: What is Django?
  A: High-level Python web framework.
- Q: Explain MTV architecture.
  A: Model-Template-View pattern.
- Q: What are migrations?
  A: Database schema changes managed by Django.
- Q: What is Django ORM?
  A: Object-relational mapper to query databases.

7. Database & SQL

- Q: What is a primary key?
  A: Unique identifier for table records.
- Q: What is a foreign key?
  A: Field linking two tables.
- Q: What is JOIN?
  A: Combines rows from multiple tables.
- Q: What is normalization?
  A: Process of organizing data to reduce redundancy.

8. Git & Deployment

- Q: What is Git?
  A: Version control system.
- Q: Difference between git pull and git fetch?
  A: Pull fetches and merges; fetch only downloads changes.
- Q: What is branching?
  A: Creating separate development lines.
- Q: How to resolve merge conflict?
  A: Manually edit conflicting files then commit.

9. HR Questions

- Q: Tell me about yourself.
  A: Brief intro, skills, projects, goals.
- Q: Explain one project deeply.
  A: Discuss problem, tech stack, challenges, results.
- Q: Why should we hire you?
  A: Skills, dedication, ability to learn and solve problems.
- Q: What are your strengths?
  A: Problem-solving, consistency, adaptability.